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LXXVIII. *Remanent Magnetism of some Dolerites, Basalts and Volcanic Tuffs from Tasmania*

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ABSTRACT

Magnetic measurements have been made on a number of samples of Tasmanian dolerites, basalts, and tuffs, varying in age from the Triassic to the early Tertiary, which were obtained by diamond drill coring. The magnetic dips of all these rocks were found to be very considerably greater than the present dip in Tasmania, and this result is believed to furnish strong evidence to support the hypothesis that a relative movement has occurred between the land mass of Tasmania and the geographical poles since the early part of the Tertiary period. Magnetic reversals were found to occur over certain ranges of depth in both the dolerites and the tuffs. The possible implications of these reversals are discussed in detail.

§ 1. INTRODUCTION

WE have recently investigated the remanent magnetization of a number of samples of Tasmanian rocks obtained by diamond drill coring. The specimens, all of which were intrusive or volcanic in origin, were obtained from bore holes in the Great Lake area, and were supplied to one of us (J.C.J.) by courtesy of the Hydroelectric Commission of Tasmania. In the region concerned a number of sills of quartz dolerite (Tholeiite) occur.‡ These sills, some of which have a thickness greater than 1000 feet, are believed to be Jurassic or Cretaceous, and to have an age of 90 to 120 million years. A number of samples from different sill complexes were

† Communicated by Dr. J. A. Clegg.

‡ These sills were intruded into Upper Triassic (Rhaetic) sediments. Subsequently, there was a long interval of erosion and peneplanation during which the roof rocks of most of the sills were stripped off and a mature lateritized peneplain was developed in the dolerites, which are more than ordinarily resistant to erosion. Subsequently, again, there was an epoch of block faulting and, later, Tertiary sediments and basalts appear, together with some limestones which are certainly Lower Miocene and possibly Oligocene. Allowing for the long time involved in peneplanation, it is considered that the possible time range of the dolerites is from late Rhaetic to Lower Cretaceous.

taken from four bore holes and were shipped to London for measurement at various times. The cores, which were $\frac{7}{8}$ in. in diameter, were cut into cylindrical specimens approximately $\frac{7}{8}$ in. in length, and the direction and intensity of magnetization of each specimen was measured using an astatic magnetometer in the way described by Blackett (1952). No attempt had been made during the drilling operations to orient the cores horizontally so that no information could be obtained as the the magnetic declination of the rocks, but since there was good reason to believe that the bore holes remained vertical throughout their entire length, the dip measurements can be taken to be reliable. The individual lengths of core were supplied with markings to distinguish between their upper and lower ends, so that it was also possible to find the sense of the magnetic dip.

Of the four cores examined, the first, which was distinguished by the code number 5001, was taken from a tholeiite sill believed to be 2000 feet thick, and specimens were obtained at intervals of approximately 50 feet down to a depth of 1000 feet. The second core, number 5002, was taken from what is believed to be another portion of the same sill as 5001. It yielded specimens of dolerite down to a depth of 146 feet, and passed below this level into a volcanic tuff which will be described later. In the remaining two cores, numbers 8102 and 8103, the sills were overlaid by 500 feet of tertiary basalt lava and samples of dolerite were obtained from depths of 500 to 700 feet.

Measurements were made initially on the doleritic sills, and were extended later to the basalts and the tuffs.

§ 2. Measurements on Dolerites

In all a total of 57 cylinders of dolerite were cut from cores 5001, 5002, and 8102. The first set of samples to be examined were from core 5001 and were taken from depths ranging from 1 to 947 feet. A detailed account of the results obtained for this core has already been given by Jaeger and Joplin (1955) and will not be repeated here. Briefly, the most significant feature was the consistent steepness of the magnetic dips, only one specimen having a direction of magnetization deviating by more than 7° from the vertical. It was also found that while the majority of the cylinders were magnetized normally (i.e. upwards), a proportion amounting to about one-fifth of the total number had reverse magnetization. These findings have since been confirmed by a further series of measurements made on the same core in Canberra by Jaeger and Joplin (1955) who, on combining the two sets of results, and giving consideration to the possibility of errors in the marking of the cores, conclude that the upper 700 feet of this particular sill is magnetized normally, while the lower part is reversed.

Following the measurements on this core, two further series of dolerite samples were obtained from cores 5002 and 8102. The results, which have not previously been published, are tabulated below.

These measurements agree very closely with those described previously for core 5001. The only specimen with a dip shallower than 75° was taken from the uppermost part of core 5002, and may well have been affected by lightning or other surface phenomena. In all, the mean value obtained for the inclination form the measurements made on all three cores is 85·5°. This is significantly greater than the present magnetic dip in Tasmania, which has a value of 70°, while the dip corresponding to an axial dipole field is 60°. The measurements therefore suggest that a relative movement has occurred between the land mass and the geographical poles at some time since the Jurassic period.

Table 1. Core 5002

Depth (feet)	Dip (+ve dip up)	Intensity ($\times 10^3$c.g.s.u.)
8	−44°	2·7
13	+86	0·91
30	+84	1·05
49	+86	0·27
71	+80	0·90
92	+82	0·38
125	+84	0·50
152	+84	0·031

Table 2. Core 8102

Depth (feet)	Dip (+ve dip up)	Intensity ($\times 10^3$c.g.s.u.)
500	+90°	4·93
518	+89	2·92
540	+83	5·20
562	+88	6·96
575	+75	7·67
598	+88	6·29
615	+89	9·20
636	+87	7·40

With regard to the magnetic reversal found at a depth of some 700 feet in the first core (no. 5001), it is possible that the rocks of the upper and lower parts of the site were injected at different times and that the Earth's field underwent a reversal in the intervening period. On the other hand, a very careful search for any geological evidence of multiple intrusion has been made and none has been found : it is considered that the sill was emplaced in a single act of intrusion. A second possibility is that the Earth's field reversed during the cooling of the sill : if this were the case it must have done so in an extremely short time. It can be calculated

that for a sill 2000 feet thick of rock with a melting range of 1000–800°C, latent heat of solidification 100 cal/gm, thermal conductivity 0·005, specific heat 0·25, and density 2·8 (units calorie and c.g.s., °C) intruded at 1000°C into country rock with the same thermal properties, it will take about 2200 years for a Curie point of 500°C to progress from the contact to the centre of the sill and about 150 years for it to move between the distances of 650 and 750 feet from the contact. Since the region above 650 feet appears to be normal, and that below 750 feet to be reversed, it seems that if a reversal of the Earth's field were ' trapped ' in the sill in this way, it must have taken place in the improbably short time of less than 150 years. Finally, of course, there remains the possibility that the apparent reversal is due to some physio-chemical effect in the rocks themselves.

§ 3. Measurements on Basalts

In addition to the study of the dolerites described above, other sections of the cores were also examined. In particular, a limited number of samples of the tertiary basalt overlying the dolerite were obtained from borings 8102 and 8103. Both these holes penetrate through a number of flows and the cores consist of massive portions interspersed with highly vesicular and much altered material. Measurements were made on eight specimens taken from the massive sections, and the results are tabulated below.

Table 3. Cores 8102 and 8103. Massive Basalts

Core No.	Depth (feet)	Dip (+ve dip up)	Intensity ($\times 10^3$c.g.s.u.)
8103	21	+82°	12·5
8103	155	+84	9·9
8103	191	−85	4·6
8103	285	+80	5·7
8103	360	+89	20·8
8102	416	+83	9·3
8102	444	+85	15·9
8102	469	+77	5·4

Samples taken from the vesicular layers between the massive portions had a much lower magnetic intensity and more widely scattered directions of magnetization, and the results for these sections have not been included in the table.

Despite the small number of specimens measured, it is evident that the magnetic dips correspond closely to those of the older tholeiite sills. There appears to be a single case of magnetic reversal at a depth of 191 feet, but with the limited amount of material available for measurement, the possibility of this being spurious, due to an error in marking one of the core sections, cannot be precluded.

§ 4. Measurements on Volcanic Tuffs

Of greater interest were the results of the investigation of the tuffs underlying the tholeiite sill in core number 5002, which are probably Triassic in age. They are greyish white in colour and have a rather coarse texture, containing abundant grains of quartz and felspar set in a friable white matrix. Although they have the characteristic features of volcanic tuffs, their precise origin is highly speculative, but the fact that a proportion of the grains are rounded suggest that they may have been transported over considerable distances, and the abundance of clay present, indicates that they were probably deposited in water. We first measured three specimens of these rocks, taken from 175 feet, 185 feet and 195 feet below the present Earth's surface. They were considerably less strongly magnetized than the dolerites and basalts, having a mean magnetic intensity of $4{\cdot}8 \times 10^{-6}$ c.g.s. units, but their directions of polarization again lay close to the vertical. It was found, however, that while the first two specimens were magnetized in a normal upward direction, the third and lowest had a downward dip. This suggested the presence of a magnetic reversal at some depth between 185 feet and 195 feet, and since a fairly complete length of core was available, we decided to investigate the variation of magnetization over this range of depths in greater detail. We, therefore, measured a number of cylinders cut at approximately one foot intervals between 174 feet and 195 feet with the following results :

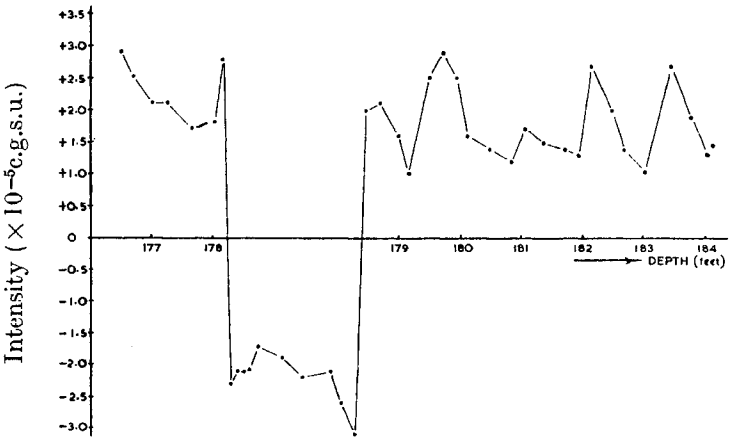
Table 4. Core 5002. Tuffs

Depth (feet) below surface	Dip (+ve dip upwards)	Intensity of Magnetization ($\times 10^6$ c.g.s.u.)
174·5	+84°	17·3
175	+78	9·0
175·5	+85	19·9
176·5	+84	6·8
177·5	+83	6·8
178·5	−80	8·2
179·5	+81	15·7
180·5	+79	8·9
181·5	+84	3·8
185	+80	4·0
186	+81	4·4
187	+79	14·0
188	+85	8·0
189	+82	5·6
190	+78	15·6
191	+81	2·6
195	−85	0·87

It can be seen from the table that in addition to the reversely magnetized cylinder obtained previously from a depth of 195 feet, a second case of downward dip occurs at 178·5 feet. Now, in the latter instance, the

particular specimen concerned had been taken from a continuous section of core, approximately 5 feet in extent, which had been carefully marked over the whole of its length before cutting in the laboratory. The normally magnetized specimens found immediately above and below, at depths of 177·5 feet and 179·5 feet, were also from the same 5 foot section, so that there was no possibility in this case of the reversal being spurious. In order to examine the variation of inclination with depth over the particular region concerned in greater detail, the whole of the 5 foot length was cut into a continuous sequence of thin disks, which were measured separately. We found the inverse magnetization to be confined to a single 5 in. length of core, around 178 feet 4 inches, throughout which the dip had a mean value of 83°, and was directed downwards: the remainder of the section, above and below this, had a mean dip of 82° but was magnetized in an upward sense. The transition from downward to upward polarization at the upper and lower extremities of this 5 in. portion was very sharp, and there was no indication of any continuous systematic variation in the direction of magnetization in passing from the reversed to the normal regions. In fig. 1 the vertical component

Fig. 1



Scale of depth over reversed 5 in. (178 ft. 1½ in.–178 ft. 8½ in.) increased by factor 3.

of the intensity of magnetization is plotted as a function of depth. There is no significant change in intensity close to the ends of the reversed portion, and the specimens showing reverse and normal polarization are, on the whole, equally strongly magnetized. It is evident, therefore that over a range of depths extending from 178 feet 1·5 inches to 178 feet 6·5 inches, the magnetic polarization undergoes a total inversion, and that the transition occurs in a space of a few millimeters, corresponding to the width of a saw cut.

We subjected a number of specimens to a series of demagnetization and other tests. The demagnetization curves for steady and alternating

fields were similar for both normal and reversed specimens, both of which had a natural coercive force of approximately 300 oersteds, and were completely demagnetized in a 50 cycle per second alternating field of about 700 oersteds. In a steady field, of the order of 10 oersteds, there was a time delay effect similar to that described previously by Clegg *et al.* (1954). The rate of rise of magnetization with time of application of field, and the rate of decay after removal of the field were the same for the reversed and normal rocks. We saturated a number of specimens of both kinds in fields of the order of 4000 oersteds, and again we found no difference in behaviour: there was a linear relationship between saturated intensity, J_s, natural remanent magnetization, J_n, the ratio J_s/J_n having a mean value of 15 for normal and reversed samples alike. The remanent coercive force after saturation was the same in both cases.

We are indebted to Mr. J. H. Leng who measured the thermal demagnetization of the rocks, and made an x-ray analysis of the separated magnetic fraction. The demagnetization curves obtained by heating normal and reversed specimens in zero magnetic field were in good agreement, and both were found to have a Curie temperature of 380°. This corresponds well with the Curie point of pyrrhotite (Fe_7S_8), and subsequent x-ray examination of the magnetic component showed it to have the expected hexagonal structure, with a lattice spacing approximating closely to that of the pyrrhotite crystal.

Reviewing the foreging results we conclude that these tuffs contain a single ferromagnetic component, with the crystal structure and magnetic behaviour of pyrrhotite, and that by none of the tests which we applied was it possible to distinguish any difference in the magnetic properties of the normal and reversed specimens.

§ 5. Discussion of Results Obtained for Volcanic Tuffs

This magnetic reversal, occurring over a narrow range of depths in the volcanic tuffs, poses a number of interesting geophysical problems. The issue is complicated by the uncertainty as to the mechanism whereby the rocks were originally magnetized. They could evidently have acquired their present polarization in one of three ways, viz.:

(i) by orientation of the magnetic particles before consolidation,

(ii) by heating of the magnetic fraction above the Curie temperature during the intrusion of the overlying dolerite sills, or

(iii) by some more subtle physical process at an unspecified intermediate period.

These three possibilities will be considered separately.

(i) *Magnetization before Consolidation*

If the rocks had been produced by a process of slow sedimentation in calm water we might have supposed the reversely polarized section to have been formed, and acquired its magnetization, during a period when

the Earth's field was reversed. But these tuffs were produced as a result of volcanic activity, and were laid down rapidly. To account for the result in terms of a change in direction of the Earth's field during deposition it would therefore be necessary to assume that the 5 in. layer showing inverse polarization, and the normally magnetized layers immediately above and immediately below, were the product of three separate volcanic eruptions, between each of which the field underwent a reversal. Now the appearance of this particular portion of the core suggests strongly that it was the product of continuous uninterrupted deposition. The whole of the 5 foot length, including the reversed portion, is uniform in mineral composition and grain size, and there is no trace of bedding planes such as would be expected if considerable intervals of time had elapsed between the deposition of different layers.

It is also possible to explain a magnetic reversal such as this in terms of relative movement between different parts of the sedimentary mass shortly after deposition. It is well known that a recently formed and unconsolidated sediment which has been laid down on a sloping surface may be subject to a considerable degree of slumping, and that this is often so pronounced that certain portions become completely inverted. Now Irving (1954) has shown that the magnetization of some slumped beds is exactly parallel in direction to that in neighbouring undeformed layers, whereas in other slumped beds the magnetic directions are widely scattered, sometimes randomly, suggesting that the magnetization may be fixed in an unconsolidated sediment prior to the deformation. In this latter case, if slumping took the form of a straight inversion, a reversal of magnetization would result. However, there is no sign of any such movements having taken place in these rocks and, even if they had occurred, it would be strange if slumping, which is regarded by geologists as a highly complicated process, should produce an exact 180° inversion without some intermediate directions as well.

(ii) *Magnetization by Heating*

Rejecting tentatively the hypotheses of changes in the Earth's field during deposition, and of slumping, there is an alternative possibility that the present magnetization of these rocks may be due to heating of the tuffs at the time when the overlying dolerite was intruded. Calculations for an intrusion with the physical properties described earlier (see § 2) show that the country rock, whose distance from the contact is less than 0·4 of the thickness of the sill, will be heated above 380°c (the observed Curie point of the present magnetic material). In the present instance there is now 140 feet of dolerite above the tuff, so that it may be expected that the latter was heated above its Curie temperature to within a distance of 56 feet below the contact. The distance may be greater than this since some of the original dolerite has been removed by weathering. The length of core in which this reversal occurs was obtained from a depth of only 30 feet below the floor of the sill, so

that it is possible that it acquired a thermoremanent magnetization at the time of injection. Indeed, since there is carbonaceous matter, and so probably sulphur, present in the tuff, it is possible that the pyrrhotite itself was actually formed during heating by the intrusion.

The observed magnetic reversal could evidently have occurred as a result of this heating. It is well known that Néel (1952) has shown on theoretical grounds how it is possible for an igneous rock to acquire a magnetization in a direction opposite to that of the Earth's field at the time of cooling. He suggested four mechanisms whereby such an effect might occur, and his predictions have been brilliantly justified by Nagata's (1953) discovery of self reversing lavas and by Gorter's (1954) synthesis of self reversing ferrites. If, however, such an effect was responsible for the observed reversal in these tuffs, there must have been some physical difference between the directly and inversely magnetized rocks at the time of cooling. That we have been unable to detect any difference in behaviour between the normal and reversed specimens, by our laboratory measurements, may be because the differences which once existed have been obliterated. Alternatively, it may be that the differences still exist but that our tests were not sufficiently sensitive to detect them.

(iii) *Magnetization by some Long Period Process*

An alternative possibility is that the whole rock mass became magnetized uniformly, either before consolidation or by subsequent heating, but that owing to some physical or chemical process which occurred at a later period, the direction of magnetization was changed over a certain range of depths. Changes such as this could take place in either of two ways: they may be due to long period relaxation effects of the kind suggested by Néel (1950, 1951) and detected in igneous rocks by Buddington and Balsley (1954) and by Asami (1954), which would cause the direction of magnetization in the rock to follow the Earth's magnetic field over long intervals of time; or they may be caused by a modification of one of the Néel self-reversing mechanisms already mentioned in a way similar to that described by Graham (1953). These alternatives will be considered separately.

If the reversal is to be explained in the first of the ways suggested, we must assume that the direction of magnetization was initially uniform throughout the whole rock mass, but that owing to long period magnetic instability, it tended to follow the variations in the Earth's field throughout subsequent geological ages. It has been shown by Néel (1950, 1951) that the relaxation times for such long period phenomena depend very critically on the grain size and on the composition of the magnetic minerals present in the rock, so that it is possible, owing to very small differences in constitution, for different sections of the same rock mass to follow the changes in the external field with vastly different time lags, and therefore to have different polarizations at the present time.

The effect of this must be considered in relation to the period of secular variation of the Earth's field, which is of the order of a thousand years, and to the average time interval elapsing between reversals of the Earth's dipole field. If we accept the estimates of Hospers (1954) and of Roche (1950), and take the latter to be of the order of a million years, we must suppose many such reversals to have occurred since these rocks were formed in Triassic times. We will assume for the purpose of the present argument that the field was reversed at the time of formation, and that while the reversely polarized section has relaxation time large in comparison with a million years, and therefore gives a true indication of the magnetic dip at that time, the normal specimens have suffered subsequent changes due to greater instability. We can examine two particular possibilities :

(*a*) If the magnetic relaxation time of the unstable rocks were small in comparison with a thousand years their magnetic directions would lie along the present Earth's field, in which case the dips would have a value of 70°. Since the measured dip (82°) of the normally polarized rocks is significantly greater than this we can discard the possibility.

(*b*) If the relaxation time were considerably longer than a thousand years, but less than a million years, we would expect the polarizations to correspond closely to that of the present axial dipole field, in which case the dips would be of the order of 60°. But if the measured dip of 82° is in fact related to the direction of the dipole field at some time in the past, this would imply that the latitude of Tasmania must then have been 74° South. Thus, if the present magnetic directions of the normally magnetized rocks have been acquired by a relaxation mechanism of this time period, we must assume the postulated Northward movement of Tasmania to have occurred since the most recent reversal of the Earth's field—conceivably within the last million years.

An alternative possibility is that the rocks may have been initially unstable but that the stability has increased at some time in the past. If, for example, the relaxation time immediately after formation was of the order of several thousand years, and if some subsequent physical or chemical change in the magnetic material caused it to increase to many millions of years, the magnetization would become 'frozen' in the direction of the dipole field at the time when the change took place. Such changes need not necessarily occur simultaneously throughout the whole of the rock mass, and if the times of freezing of different parts extened over a number of field reversals we might expect different sections to have diametrically opposite polarizations.

Owing to the great experimental difficulties encountered in obtaining reliable information concerning these long period relaxation phenomena, and the paucity of the present knowledge on the subject, it is not feasible to make any clear estimate of the extent to which the remanent magnetism of these rocks can be attributed to such causes.

Turning now to the possibility that the inversely polarized rocks were magnetized by a long period self reversing process which may have taken

place at some time since they acquired their original remanent magnetism, Graham (1953) has put forward a plausible suggestion, based on a consideration of one of Néel's (1952) mechanisms, to show how it might be possible for certain red sandstones to suffer reversal as a result of chemical action after deposition ; and it is possible to visualise other variants of the Néel processes which might cause self reversals to occur in sedimentary rocks. We cannot preclude the possibility of some such magnetic change having occurred in the volcanic tuffs under consideration here, and our results could be explained on the assumption that the rocks were laid down at a time when the Earth's field had the same sense as it has today, and that they became uniformly magnetized in that direction, but that at some later period a magnetic self reversal occurred throughout a particular layer 5 in. in depth. If such were the case, there must evidently have been a difference between the magnetic condition of the layer in which the change took place and the layers above and below which retained their original polarity. As pointed out previously, however, we have been unable to detect any such difference.

Finally, therefore, it seems impossible, on the basis of the present experimental measurements, to form any definite conclusion as to how the observed magnetic reversal occurred in these tuffs. The results, however, do serve to underline the uncertainties attending any discussion of field reversal. There is strong presumptive evidence to support the belief that the Earth's magnetic field has reversed at times during past geological ages. The best of this evidence is furnished by the experiments carried out by Hospers (1954) by Nagata (1953), by Kato *et al.* (1955), and more recently by Leng (1956), on the magnetization of baked rocks surrounding igneous dykes. In these instances, the geological conditions were such that there could be little doubt as to how and when the magnetization was acquired. But in the present case the magnetic history of the material is by no means certain, and it has been shown that there are at least two possible explanations, based on the hypotheses of slumping, and of a self reversing mechanism, which could account for the observed results without stipulating any change in the Earth's dipole field in past ages. It may be that with the development of more searching laboratory tests in the future, some of the present obscurities will disappear. In the meantime every case of magnetic reversal in rocks must be treated on its own merits, and its implications must be considered with reserve.

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